

Exploratory Data Analysis (EDA) Summary Report Template

1. Introduction

The primary objective is to analyze customer data and predict delinquency risks using AI-driven techniques.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

The primary target variable is Delinquent_Account (binary: 0 or 1), making this dataset well-suited for binary classification modeling and risk segmentation.

Key dataset attributes:

- **Number of records:** 500 customers , 19 columns

- **Key variables :** Datatype [19]

Binary Categorical : Delinquent_Account [1]

- Binary flag indicating if customer has a delinquent account (0 = No, 1 = Yes)

Ordinal Categorical : Months_1 to Months_6 [6]

- Payment status : On-time, Late, Missed

Categorical : Customer_ID, Location, Employment_Status, Credit_Card_Type [4]

- *Customer_ID* : Unique customer identifier

- *Location* : City of residence (5 cities = LA, Phoenix, Chicago, Houston, New York)

- *Employment_Status* : Employment type (Employed, Self-employed, Unemployed, Retired)

- *Credit_Card_Type* : Type of credit card (Student, Standard, Gold, Platinum, Business)

Numerical : Income, Credit_Utilization, Loan_Balance, Debt_to_Income_Ratio, Age, Credit_Score, Missed_Payments, Account_Tenure [8]

- *Income* : Annual income

- *Credit_Utilization* : Credit utilization ratio, Fraction of available credit in use , Accounts with utilization rates approaching or exceeding 1.0 are key segments for immediate risk review.

- *Loan_Balance* : Outstanding loan balance (the total amount of money that a

borrower still owes to the lender at a specific point in time)

- *Debt_to_Income_Ratio* : Debt as a fraction of income (0.10–0.55)

- *Age* : Customer age in years (18–74)

- *Credit_Score* : Credit score rating (301–847)

- *Missed_Payments* : Number of missed payments (0–6)

- *Account_Tenure* : Years the account has been open (0–19)

Column Summary & Key Patterns

1. *Customer_ID* (500/500 non-null)

Unique identifier for each customer

No missing values

2. *Age* (500/500 non-null)

Range: 18 - 74 years

Distribution: Broad spread across working age

Common Pattern: Majority between 30-60 years

3. *Income* (461/500 non-null, 39 missing)

Missing: 7.8% of records

Range: ~15,400 – ~200,000

Pattern: Wide variation; many values appear rounded.

4. *Credit_Score* (498/500 non-null, 2 missing)

Missing: 0.4% of records

Range: 301 – 847

Pattern: Bimodal distribution with clusters at lower and higher score ranges.

5. *Credit_Utilization* (500/500 non-null)

Range: 0.05 - 1.025

Pattern: Some values capped at 0.05 (minimum) and exceed 1.0 (over-utilization)

Common around 0.3-0.7

6. *Missed_Payments* (500/500 non-null)

Range: 0 - 6

Pattern: Count of missed payments (higher = riskier)

7. *Delinquent_Account* (500/500 non-null) - Target Variable

Distribution:

0 (No delinquency): 84% (420 customers)
1 (Has delinquency): 16% (80 customers) Imbalanced: Minority class is delinquency cases

8. Loan_Balance (471/500 non-null, 29 missing))

Missing: 5.8% of records
Range: 612 – 99,620.

9. Debt_to_Income_Ratio (500/500 non-null)

Range: 0.1 - 0.549
Pattern: Many values at exactly 0.1 (floor/capped values)
Common range: 0.2 - 0.4

10. Employment_Status (500/500 non-null)

Categories (frequency pattern): employed/Employed/EMP - Most common (multiple variants), Self-employed, Unemployed, Retired
Note: Inconsistent capitalization (e.g., "employed" vs "Employed" vs "EMP")

11. Account_Tenure (500/500 non-null)

Range: 0 - 19 years
Pattern: Fairly uniform distribution; many customers with 10+ years

12. Credit_Card_Type (500/500 non-null)

Categories: Student (most common), Standard, Gold, Platinum, Business

13. Location (500/500 non-null)

Cities: New York, Los Angeles, Chicago, Houston, Phoenix
Distribution: Relatively balanced across 5 cities

14. Month_1 to Month_6 (500/500 non-null)

Statuses: On-time, Late, Missed
Pattern:
Most customers show at least some late/missed payments
Month_6 has more "On-time" than earlier months (possible end-of-period effect)

Predictive Modeling (Delinquency Prediction)

Category	Features	Purpose
Demographic	Age, Location, Employment_Status	Customer profile

Financial Health	Income, Credit_Score, DTI, Loan_Balance	Financial stability
Credit Behavior	Credit_Utilization, Missed_Payments, Card_Type	Spending/history pattern
Relationship	Account_Tenure	Loyalty/experience
Payment History	Month_1 to Month_6 (6 variables)	Recent payment behavior
Target	Delinquent_Account	What we're predicting

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key **missing data** findings:

Out of 500 total records, missing values are concentrated in three columns:

- Income (39 missing values, or 7.8%)
- Loan_Balance (29 missing values, or 5.8%)
- Credit_Score (2 missing values, or 0.4%)

Inconsistency was found in the Employment_Status column. The categories 'Employed', 'employed', and 'EMP' represent the same status but are formatted differently.

Actionable Step: Before applying any imputation strategy, these categorical values must be standardized (e.g., converting all variations to 'Employed') to ensure that any relationship between employment and missing numeric fields is modeled correctly.

1. Strategy for Low-Missingness Features (*Credit_Score*)

Recommended Strategy: Median Imputation. For features with extremely low missingness like Credit_Score (0.4% missing), simple statistical imputation is the industry standard because it is computationally efficient and will not noticeably distort the column's distribution.

Justification: The Credit_Score column exhibits a highly symmetric distribution with a skewness near zero, a mean of 577.7, and a median of 586.0. Utilizing the median value

of 586 to fill the two missing cells is robust, safe, and prevents outliers from skewing the data.

2. Strategy for Moderate-Missingness Features (*Income & Loan_Balance*)

Recommended Strategy: K-Nearest Neighbors (KNN) Imputation or Group-Based Median Imputation. Because Income (7.8% missing) and Loan_Balance (5.8% missing) have higher rates of missingness, applying a single global mean or median would artificially compress the variance and weaken the predictive power of these variables for delinquency modeling.

Justification: Industry best practice for moderate missingness is KNN Imputation (typically using $k=5$ neighbors), which estimates the missing value based on the profiles of the most similar customers (matching on attributes like Age, Location, and Credit_Score).

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key findings:

Key Risk Indicators Identified During EDA :

1. Missed_Payments = Past behavior predicts future behavior
2. Credit_Utilization = High utilization = living beyond means
3. Credit_Score = Reflects overall creditworthiness
4. Debt_to_Income_Ratio = High DTI = too much debt relative to income
5. Account_Tenure = Longer tenure = more stable customer
6. Income = Helps but not decisive

What matters more? = Payment history, credit score, and how much credit someone uses

Income has a weak but statistically significant negative correlation with delinquency risk. However, factors like payment history, credit utilization, and credit score are much stronger predictors. When assessing risk, focus on behavior patterns first, income second.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. This section documents AI-generated insights and the prompts used to obtain results.

AI prompts used: (Deepseek, Gemini)

- Analyze this dataset and provide a summary of key columns, including common patterns and missing values
- Identify missing values in this dataset and recommend the best imputation strategy based on industry best practices.
- Analyze the correlation between customer income and delinquency risk, summarizing key findings in simple terms.
- Analyze trends in late payments and identify the top 3 risk factors associated with delinquency.

6. Conclusion & Next Steps

Further investigation

Priority	Action	Implementation
1.	Monitor Missed_Payments (Corr: +0.48)	Flag customers with ≥ 3 missed payments for early intervention
2.	Track Credit_Utilization (Corr: +0.33)	Alert when utilization exceeds 70%
3.	Review Credit_Score Drops (Corr: -0.30)	Significant drops (50+ points) should trigger review
4.	Combine All Three	Create risk score: High if (Missed ≥ 3 OR Util $>70\%$ OR Score <450)

The identified risk indicators should be incorporated into the predictive modelling framework rather than combined into an arbitrary manual risk score. Logistic Regression, Random Forest, and XGBoost will be compared using stratified cross-validation, with the final classification threshold selected based on the business trade-off between false positives and false negatives.